Student's Name: **Enrolment No:** **Date:**	rishihood university

End Semester Examination – May 2026

Course Code: CSA331
Course Title: Advanced Discrete Mathematics

Programme: B.Tech CS & AI **Semester: IV**

Duration: 3 Hours **Max. Marks: 40**

Instructions:

1. This paper contains two sections — Section A and a Bonus Section.
2. Section A contains 11 questions. Attempt any 8 questions. Each question carries 5 marks.
3. Bonus Section contains 5 questions. Each question carries 1 mark. Marks achieved in the Bonus Section will be added to your total marks (capped at 40).
4. Show all calculation steps and provide reasoning where necessary.
5. You may leave answers in factorial or combination form where appropriate.
6. Do not discuss anything with other students during the examination.
7. Scientific calculators are not permitted.

Section - A

Attempt any 8 questions.

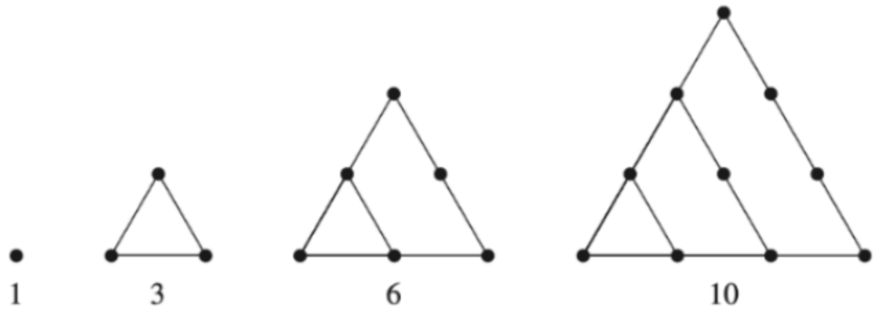
1. Prove that the following are tautologies using truth tables or logical equivalence laws:
   a. $[B' \wedge (A \rightarrow B)] \rightarrow A'$
   b. $[(A \rightarrow B) \wedge A] \rightarrow B$
   c. $(A \vee B) \wedge A' \rightarrow B$

2. Verify that $(\mathbb{Z}_8, +_8)$ is a cyclic group. Find all its generators.

3. Prove that there is no smallest positive rational number.

4. Early members of the Pythagorean Society defined figurate numbers as the number of dots in certain geometric configurations.

   The first few *triangular* numbers are 1, 3, 6, and 10:



   Find and solve a recurrence relation for the nth triangular number.

   Note: The *n*th *triangular* number has *n* dots on a side (see the triangles above).

5. The Tower of Hanoi is a classic mathematical puzzle involving three rods (A, B, and C) and n disks of different sizes. Initially, all disks are stacked on rod A in decreasing order of diameter — the largest disk at the bottom and the smallest at the top.
   Goal: Move the entire stack from rod A to rod C while strictly following these rules:
   - Move only one disk at a time.
   - At each step, you can take the top disk from any rod and place it on another rod.
   - A disk can only be moved if it is the topmost disk of a rod.
   - No larger disk may be placed on top of a smaller disk.

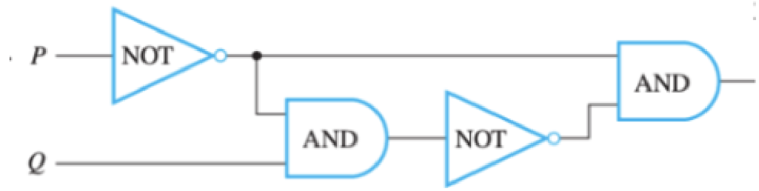
   Using diagrams, explain your approach to solving this problem. Hence,
   a. Formulate the recurrence relation for the minimum number of moves T(n).
   b. Derive the closed form of the recurrence relation and verify it for n=3.

6. Using Cantor's Diagonalization argument, prove that the set of real numbers in (0,1) is uncountable.

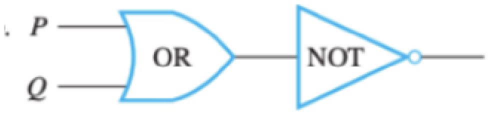
7. Find the ordinary generating function and its closed form for the sequence $a_n = 3^n$.

8. Define a ring and a field with all axioms. Show that $\mathbb{Z}_7$ under addition and multiplication modulo 7 forms a field.

9. Show that the following two logic circuits are equivalent by proving that they produce identical outputs for all input combinations.
    a.



    b.



10. Solve the following
    a. Al, Bob, Cam, Dan, and Euclid are all members of the social networking website Facebook. The site allows members to be "friends" with each other. It turns out that Al and Cam are friends, as are Bob and Dan. Euclid is friends with everyone. Represent this situation with a graph.
    b. Among a group of 5 friends, is it possible for everyone to be friends with
        i. Exactly 3 people.
        ii. Exactly 4 people.

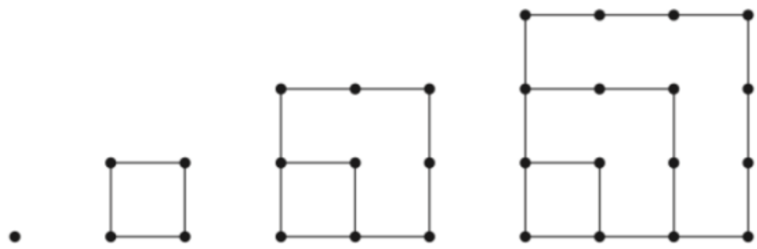
11. Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 1$, $a_1 = 4$ using the characteristic root method. Verify your solution for n = 3.

Bonus Section

1. Prove the following identity for the Fibonacci sequence:
   $F(n + 4) = 3F(n + 2) − F(n)$ for all $n \geq 1$

2. The Peirce arrow is defined as: $P \downarrow Q \equiv \neg(P \vee Q)$. Write $P \rightarrow Q$ using Peirce arrows only.

3. The First Few Square Numbers are 1, 4, 9, and 16 (see question 4 above).



1 4 9 16

Find and solve a recurrence relation for the *n*th square number.

4. Can there be a simple graph that has n vertices all of different degrees? Justify your answer.

5. The *Online Encyclopedia of Integer Sequences* (OEIS) was originated and maintained for many years by Neil Sloane, a mathematician at AT&T who has also written several books about sequences. The OEIS Foundation now manages the database, which contains more than 200,000 sequences of integers that have been submitted and studied by many people. (See oeis.org). There is even a YouTube movie about the OEIS!
Recaman's sequence* (Find it in OEIS catalog after exam) is a recursive sequence defined as follows:

$$a(1) = 1$$

For $n > 1$,

$$a(n) = \begin{cases} a(n-1) - n \text{ if that number is positive and not already in the sequence,} \\ \quad \text{otherwise} \\ a(n-1) + n \end{cases}$$

   a. Find the first 5 terms of this sequence.
   b. Find the index of this sequence at which the following numbers appear: 10, 12, 23.

**It has been conjectured that every nonnegative integer will eventually appear in this sequence.*